

Kalshi Crypto Desk

Probabilistic auto-trading for Kalshi BTC/ETH & index contracts

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Kalshi Crypto Desk — Pitch Brief

Probabilistic auto-trading for Kalshi BTC/ETH and index hourly/15m contracts

Version Beta 4.0.60 · Self-hosted · ML-first · Settlement-native

One-liner

We built a **quant trading stack** for Kalshi crypto and index hourly/15-minute markets: LightGBM models estimate fair probability, execution only fires when **ask-edge beats fees**, and a full ops dashboard runs paper, live, and A/B trials side by side. It is not a generic rule bot — it is a **prop desk in a box** for contracts that settle on BRTI/ERTI.

The problem

Kalshi crypto and index products (hourly thresholds, range bands, 15m slots) move fast and settle on **official index references**, not exchange spot. Most traders either trade manually and miss entries, use no-code rule bots with no model of fair value, or spend months building Python, Kalshi API integration, ML, and dashboards from scratch.

There is a gap between **automation** and **edge**.

Our solution

An end-to-end system:

Layer	What it does
Predict	LightGBM on OHLCV + structure features; daily retrain + 6h isotonic recalibration
Map	Model probability → Kalshi contract picks (threshold, range, 15m slot)
Gate	Fee-adjusted ask-edge, regime filters, Kelly sizing, correlation guards, cooldowns
Execute	Paper or live via Kalshi API; cross-spread on high edge; optional bracket orders
Learn	Adaptive bucket pause/probe from trade P&L; auto-tune on passive bots
Report	Dashboard: P&L, skip reasons, live vs paper/trial compare, reconcile health

Six independent bots today: BTC hourly, BTC hourly trial, BTC 15m, ETH hourly, ETH hourly trial, ETH 15m — plus SPX/NDX hourly experiments. Each has separate state, risk cap, and settings.

How we compare to Bot for Kalshi

Bot for Kalshi (\$99/mo) is a **broad no-code platform**: visual step builder, any market, encrypted multi-user keys, templates (mean reversion, momentum, settlement yield). Excellent for **general Kalshi automation**.

We are purpose-built for probability mispricing on index-settled crypto and index products.

Dimension	Bot for Kalshi	Our stack
Core edge	Rules you define (price, time, triggers)	**ML probability vs market ask**
Markets	All Kalshi	**BTC/ETH hourly + 15m** (+ index hourly)
Settlement	You configure	**BRTI/ERTI gates** , hour/slot rollover force-close
Research loop	In-app template backtests	**Paper + live + trial bots** , performance reports
Execution depth	Brackets, stops, limits	Ask-edge, cross-spread, whipsaw guard, range caps
Who runs it	They host	**You host** (Railway); full control, no platform rent
Multi-user	Built-in	Roadmap (separate deploys today; in-app tenants later)

Analogy: Bot for Kalshi is **Zapier for Kalshi**. We are a **small quant fund's infra** for one product vertical.

Why this matters on Kalshi

- **Contracts are binary and fee-sensitive** — edge is measured in cents vs model prob, not vibes.
- **Settlement index $\neq$ exchange spot** — live entries require BRTI/ERTI; we enforce that.
- **Hourly churn is brutal** — exits-before-entries, hybrid/adaptive TP, cut-loss cooldowns, hour-cap discipline.
- **Paper lies if execution differs** — live vs paper/trial compare catches fill drift (resting limits vs cross-spread).

Traction and proof (honest stage)

- **Live trading** on personal Kalshi account with versioned deploys (Beta 4.0.x).
- **Paper and trial bots** in parallel for controlled experiments (leg-stop trial vs thesis hourly).
- **Operational maturity**: WAL SQLite, 429 circuit breaker, live reconcile, DB backups, settlement cleanup.
- **Not claiming** audited track record or guaranteed returns — research software maturing toward durable edge.

Go-live checklist: 20+ hours flat/positive hour P&L, reconcile stable, TAKE PROFIT dominates cuts, no phantom settlement rows.

Business model options

Path	Description
Personal alpha	Keep private; compound on own account
Managed accounts	Separate Railway + Kalshi key per client (works today)

Path	Description
SaaS (future)	Multi-tenant users, encrypted keys, shared strategy deploys
Licensing	Stack for sophisticated Kalshi traders who want ML + ops

Infrastructure: ~\$5–20/mo hosting vs \$99/mo generic bot platforms — you own maintenance.

Competitive moat (if productized)

- **Domain-specific ML pipeline** — 15m, hourly, 2nd-chance models with production calibration
- **Execution rule library** — tuned on real Kalshi friction, not replicable in a visual builder quickly
- **Compare infrastructure** — paper/trial/live on same clock = faster iteration
- **Index product focus** — depth on KXBTCD, KXETH15M vs shallow coverage across all markets

The ask

Collaborators: Positioning feedback, benchmark design (us vs Bot for Kalshi on same caps), ops help.

Investors: Capital for productization (multi-user, encrypted keys, onboarding) and longer live track record — not for guaranteed bot returns.

Early users (future): Hosted seats: same bots, your Kalshi key, strategy updates on every deploy.

Closing line

Bot for Kalshi helps you automate what you already believe. We help you discover and trade what the model believes — with the guardrails a Kalshi crypto desk actually needs.

Disclaimer: Prediction markets involve risk of loss. Past paper or live results do not guarantee future performance. Not financial advice.